SYSC 3303-A

Firefighting Drone Swarm System

Project Iteration 0

Group A2 - 01
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Core Parameters

Parameter	Value	Unit	Source	URL/Reference	Notes
TRAVEL TIME BETWEEN ZONES					
Maximum Speed - T40	12	m/s	DJI Agras T40	https://ag.dji.com/t40/specs	Max flight speed
Maximum Speed - T30	10	m/s	DJI Agras T30	https://ag.dji.com/t30	Max level speed specification
Maximum Speed - P100	13.8	m/s	XAG P100	https://www.xa.com/en/p100	Maximum Flight Speed
Selected Cruise Speed	**8**	**m/s**	**Calculated**	**Average of all three**	**Conservative cruise (67-80% of max)**
Takeoff Acceleration	1.5	m/s^2	Estimated from all three	Based on heavy-lift drone specs	Conservative for 43kg loaded
Landing Deceleration	1	m/s^2	Estimated from all three	Slower for safety	Standard practice
NOZZLE DOOR OPERATION					
Door Open Time	0.3	seconds	Estimated	Typical servo motor	Standard servo motor speed
Door Close Time	0.3	seconds	Estimated	Same as open	Symmetric operation
WATER/FOAM DROP RATE					
Flow Rate - T40	12	L/min	DJI Agras T40	https://ag.dji.com/t40/specs	Max Flow Rate: 6 L/min*2
Flow Rate - T30	7.2-8	L/min	DJI Agras T30	https://ag.dji.com/t30	Maximum spray flow with nozzles
Flow Rate - P100	12	L/min	XAG P100	https://www.xa.com/en/p100	Max. Operating Flow Rate
SELECTED CRUISE SPEED	**7.2**	**L/min**	**Conservative choice**	**Lowest value from all three**	**Ensures feasibility (0.12 L/s)**
Drop Time (10L - Low)	83.3	seconds	Calculated	10L ÷ (7.2/60) L/s	Pure spray time
Drop Time (20L - Moderate)	166.7	seconds	Calculated	20L ÷ (7.2/60) L/s	Pure spray time
Drop Time (30L - High)	250	seconds	Calculated	30L ÷ (7.2/60) L/s	Needs multiple drones
ACCELERATION/DECELERATION					
Takeoff Time	5.3	seconds	Calculated	8 m/s ÷ 1.5 m/s^2	Time to reach cruise speed
Takeoff Distance	21.2	meters	Calculated	½ × 1.5 × 5.3^2	Distance during acceleration
Landing Time	8	seconds	Calculated	8 m/s ÷ 1.0 m/s^2	Time to full stop
Landing Distance	32	meters	Calculated	½ × 1.0 × 8.0^2	Distance during deceleration

Travel Time Calculations

From	To	Distance (m)	Accel Time (s)	Cruise Time (s)	Decel Time (s)	Total Time (s)
Base (0,0)	Zone 1 (350,300)	461.6	5.3	51.1	8	**64.4**
Base (0,0)	Zone 2 (325,1050)	1099.4	5.3	130.4	8	**143.7**
Zone 1 (350,300)	Zone 2 (325,1050)	752	5.3	87.4	8	**100.7**
Formulas Used:	Distance = $\sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$	Cruise Distance = Total Distance - 21.2m - 32m	Cruise Time = Cruise Distance ÷ 8 m/s	Total Time = 5.3s + Cruise Time + 8.0s		

Mission Time Breakdown

Shows how all 4 required parameters combine in a real mission:

Severity	Agent (L)	Travel Example (s)	Accel/Decel (s)	Door Open (s)	Drop (s)	Door Close (s)	Total Mission (s)	
Low	10	64.4 (to Zone 1)	included	0.3	83.3	0.3	**148.3**	
Moderate	20	64.4 (to Zone 1)	included	0.3	166.7	0.3	**231.7**	
High	30	64.4 (to Zone 1)	included	0.3	250	0.3	**315.3**	Although real-world agricultural drones spray over long durations, the simulator allows these parameters to be scaled for testing. The selected values prioritize realism while remaining configurable in later iterations.

Reference Drones

Comparison of three specific drones with similar payload capacity:

Drone Model	Max Payload (kg)	Max Speed (m/s)	Tank Capacity (L)	Spray Rate (L/min)	Flight Time (min)	URL
DJI Agras T40	40	12	40	12	18-Jul	https://ag.dji.com/t40/specs
DJI Agras T30	30	10	30	7.2-8	7.8-20.5	https://ag.dji.com/t30
XAG P100	40	13.8	40	12	16-Jun	https://www.xa.com/en/p100

Why these drones:

All carry sufficient payload (10-15kg requirement met)

All are agricultural/spray drones with proven delivery systems
Specifications are publicly available and verified

References

#	Source	Model/Title	URL	Data Used	Date Accessed
1	DJI	Agras T40 Specifications	https://ag.dji.com/t40/specs	Max speed 12 m/s, spray rate 12 L/min, payload 40kg	18-Jan
2	DJI	Agras T30 Specifications	https://ag.dji.com/t30	Max speed 10 m/s, spray rate 7.2-8 L/min, comparison data	18-Jan
3	XAG	P100 Agricultural Drone	https://www.xa.com/en/p100	Max speed 13.8 m/s, spray rate 12 L/min, payload 40kg	18-Jan

Justifications

Brief explanations for the 4 key parameters:

Parameter Type	Value	Why This Value?
Travel Time	8 m/s cruise	DJI Agras T40 max speed is 12 m/s, T30 is 10 m/s, XAG P100 is 13.8 m/s. Using 8 m/s as conservative cruise speed (67-80% of max) for energy efficiency, payload stability, and control margin.
Acceleration	1.5 m/s² takeoff	Estimated from thrust-to-weight ratios of heavy-lift drones. XAG P100 and T40 have similar ratios. Conservative for safety with full payload.
Deceleration	1.0 m/s² landing	Slower than takeoff acceleration for controlled, safe descent based on drone landing procedures.
Nozzle Doors	0.3 seconds	Estimated from typical servo motor response times used in agricultural drones.

Drop Rate	7.2 L/min	DJI Agras T40 specs list 12 L/min, T30 lists 7.2-8 L/min, XAG P100 lists 12 L/min. Using conservative 7.2 L/min (0.12 L/s) for consistency; lowest value. PS: Agricultural drones are designed for sustained spraying over crops; therefore, manufacturer spray rates are specified in litres per minute. These values were converted to litres per second for simulation timing.

Analysis

Parameter Selection:

All simulation parameters were selected using conservative values taken from real world drone specifications. Choosing values must account for payload, safety margins, and operating variability, while ensuring that all timing calculations remain realistic and consistent across different drone models.

Travel Time:

The travel time table shows that acceleration and deceleration add a fixed time to every mission. For shorter distances, this fixed time makes up a large portion of total travel time, which supports modeling drone motion using acceleration, cruise, and deceleration phases instead of assuming constant speed.

Mission Time:

The mission time breakdown shows that total mission time increases with fire severity because higher severities require more water or foam. For high severity fires, the drop time is the largest part of the mission, meaning overall response time is more limited by delivery than by travel.

Reference Drones:

The reference drone table shows that maximum speed and spray rate vary across different commercial drones. Using a conservative choice for cruise speed and spray rate ensures a realistic simulation under full payload conditions.